



Create, test, and deploy a custom question answering project

COM682
Practical Lab 11

Contents

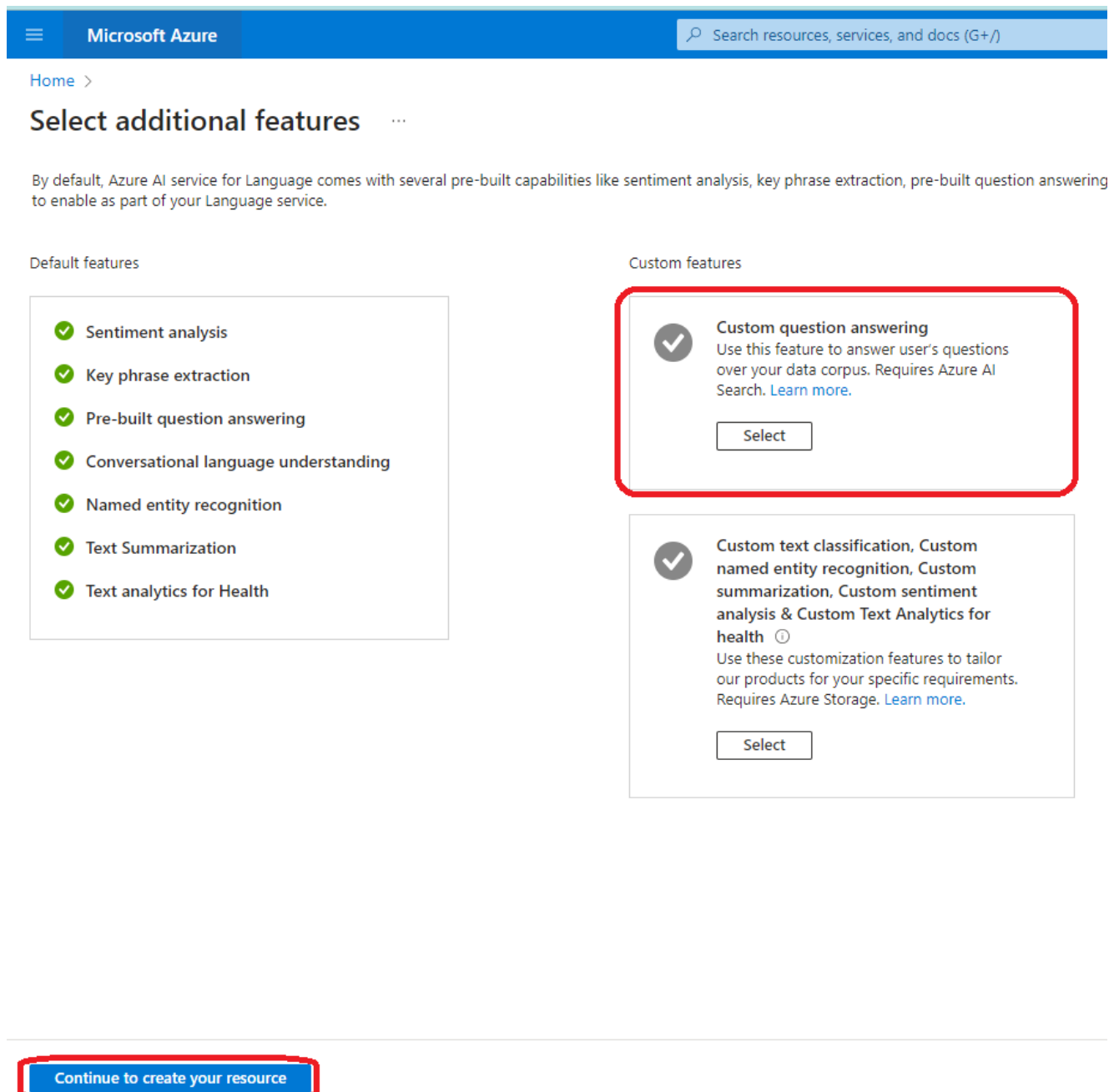
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Preamble

You can create a question answering project from your own content, such as FAQs or product manuals. This article includes an example of creating a question answering project from a product manual, to answer questions.

- If you don't have an Azure subscription, create a [free account](#) before you begin.
- A [language resource](#) with the custom question answering feature enabled.

Click on [language resource](#), it will open Your Azure Account with the following Page:



Microsoft Azure

Search resources, services, and docs (G+/)

Home >

Select additional features

By default, Azure AI service for Language comes with several pre-built capabilities like sentiment analysis, key phrase extraction, pre-built question answering to enable as part of your Language service.

Default features

- ✓ Sentiment analysis
- ✓ Key phrase extraction
- ✓ Pre-built question answering
- ✓ Conversational language understanding
- ✓ Named entity recognition
- ✓ Text Summarization
- ✓ Text analytics for Health

Custom features

- ✓ **Custom question answering**
Use this feature to answer user's questions over your data corpus. Requires Azure AI Search. [Learn more.](#)
- ✓ **Custom text classification, Custom named entity recognition, Custom summarization, Custom sentiment analysis & Custom Text Analytics for health** ⓘ
Use these customization features to tailor our products for your specific requirements. Requires Azure Storage. [Learn more.](#)

Select the Custom Question Answering and click on Continue to create your resource.

Create Language ...

[Basics](#) [Network](#) [Identity](#) [Tags](#) [Review + create](#)

Unlock insights from unstructured text using advanced natural language processing. Use sentiment analysis to find out what customers think of your brand. Find topic-relevant phrases using key phrase extraction and identify the language of the text with language detection. Detect and categorize entities in your text with named entity recognition.

[Learn more](#)

Project Details

Subscription *	Azure for Students (12ea482d-3306-4291-b560-f66da07670e7)
Resource group *	

[Create new](#)

Instance Details

Region	East US
Name *	
Pricing tier *	

[Select the free one here](#)

[View full pricing details](#)

✖ The value must not be empty.

i The free tier (F0) for this resource type is already being used by your subscription, therefore it will not appear in the dropdown below.

Custom question answering

Custom question answering lets you answer user's questions over your data corpus. You can extract questions and answers from your data, customize them and create a knowledge base. The knowledge base is stored in an Azure AI Search index in your own subscription.

[Learn more](#)

Azure search region	East US
Azure search pricing tier *	Free F (3 Indexes)

[View full Azure search pricing details](#)

Responsible AI Notice

Microsoft provides technical documentation regarding the appropriate operation applicable to this Azure AI service that is made available by Microsoft. Customer acknowledges and agrees that they have reviewed this documentation and will use this service in accordance with it.

[Responsible Use of AI documentation for Text Analytics for Health](#)

[Responsible Use of AI documentation for PII](#)

[Responsible Use of AI documentation for Language](#)

By checking this box I certify that I have reviewed and acknowledge the terms in the Responsible AI Notice. *

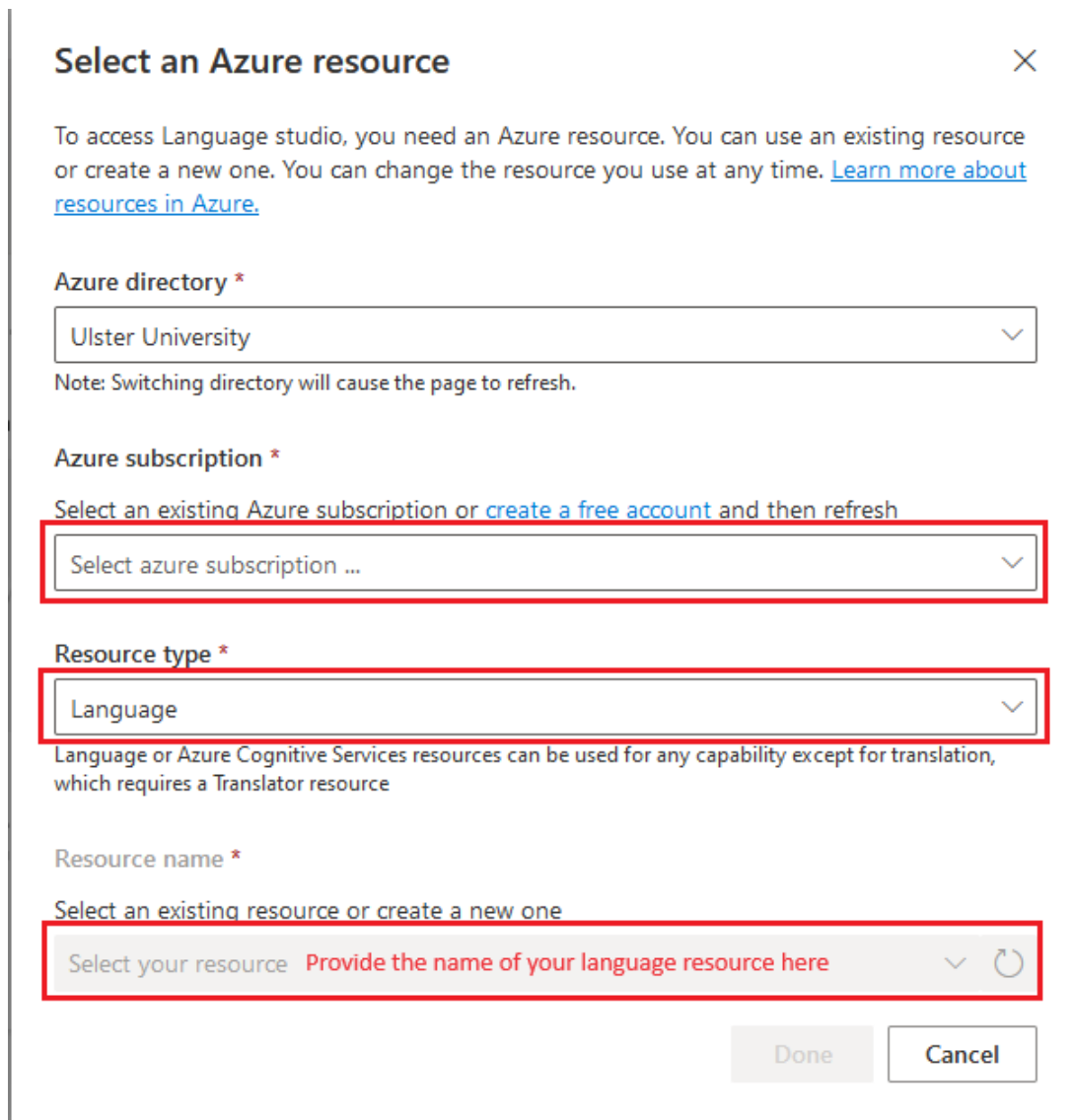
☐

[Tick this box](#)

[Previous](#) [Next](#) [Review + create](#)

Provide the required information in the previous images and then Select the review + Create tab.
Finally click the create button.

Next, Sign in to the [Language Studio](#) with your Azure credentials. Once logged in, the below page will open:



Select an Azure resource ✕

To access Language studio, you need an Azure resource. You can use an existing resource or create a new one. You can change the resource you use at any time. [Learn more about resources in Azure.](#)

Azure directory *

Ulster University ▼

Note: Switching directory will cause the page to refresh.

Azure subscription *

Select an existing Azure subscription or [create a free account](#) and then refresh

Select azure subscription ... ▼

Resource type *

Language ▼

Language or Azure Cognitive Services resources can be used for any capability except for translation, which requires a Translator resource

Resource name *

Select an existing resource or create a new one

Select your resource **Provide the name of your language resource here** ▼ ↺

Done Cancel

Provide the required information, and click on Done. A new page will open as follows:

Language Studio

Welcome to Language Studio

Recent custom projects you've worked on

You don't have any recent projects yet. Start with one of the custom capabilities to create a new project. The list of recent projects you've worked on will then appear here.

[Create new](#)

★ **Featured** | Extract information | Classify text | Understand questions and conversational language | Summarize text | Translate text

Check out some of our newest featured capabilities that we are offering in the Language Studio.



Post call transcription and analytics (Preview)

Batch transcribe call center recordings and extract valuable information such as Personal Identifiable Information (PII), sentiment, and call summary.

[Try out post call transcription](#)



Summarize information

Summarize the most important or relevant information within documental and conversational text.

[Try it out](#)



Document translation (Preview)

Batch translate documents into one or more languages either from local storage or Azure Blob Storage.

[Translate documents](#)

Learning resources

[Read the documentation](#)

Interpret natural language, classify documents, and real-time translations.

[Explore our code samples](#)

Explore our samples and discover the ways you can build with Language Studio.

[Watch a video \(coming soon\)](#)

Understand language and infuse intelligence in your apps with Language Studio.

[Microsoft Learn \(coming soon\)](#)

Discover new skills, find certifications, and advance your career in AI.

Click on the Create New, and Select Custom Question Answering as follows:

You don't have any recent projects yet. Start with one of the custom capabilities to create a new project. The list of recent projects you've worked on will then appear here.

[Create new](#)

- Conversational language understanding
Build natural language into apps, bots, and IoT devices.
- Orchestration workflow
Connect and orchestrate CLU, Custom question answering & LUIS project...
- Custom question answering**
Customize the list of questions and answers extracted from your content C...
- Custom text classification
Train a classification model to classify text using your own data.
- Custom named entity recognition
Train an extraction model to identify your domain categories using your o...
- Custom conversation summarization
Train a summarization model to create or generate summaries of conversa...
- Custom abstractive summarization
Train a summarization model to create or generate summaries of docume...

★ **Featured** | Extract information | Classify text | Understand questions and conversational language | Summarize text | Translate text

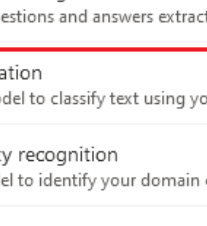
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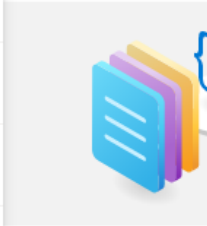
[Try out post call transcription](#)



Summarize information

Summarize the most important or relevant information within documental and conversational text.

[Try it out](#)



Document translation (Preview)

Batch translate documents into one or more languages either from local storage or Azure Blob Storage.

[Translate documents](#)

Select English and click on Next as follows:

Create a project

Choose language setting

Enter basic information

Review and finish

Choose language setting for resource LanguagestudioSUST.

Permanently set whether or not you can create projects in multiple languages using your Azure resource LanguagestudioSUST. [Learn more about projects in multiple languages and pricing.](#)

How do you want to select the language for projects in this resource?* ⓘ

☐ I want to select the language when I create a project in this resource
When creating a project in this resource you will be able to select what language the data is in. Selecting this option will incur more costs. [Learn more about pricing](#)

☒ I want to set the language for all projects created in this resource
All projects created in this resource will always use the same language for the data.

Select the language for all projects* ⓘ

English

Back

Next

Create project

Cancel

Next, provide a name and description (Optional) to your project and click Next.

Create a project

Choose language setting

Enter basic information

Review and finish

Enter basic information

Enter the basic information for your custom question answering knowledge base such as name and description.

LanguagestudioSUST

To change your resource go to [Settings](#)

Azure search resource

languagestudiosust-as2w2hjlhwppzm

To change your resource go to [Azure Search](#)

Name *

ABC

Description

Type project description

Source language * ⓘ

English

Default answer when no answer is returned * ⓘ

No answer found

Back

Next

Create project

Cancel

Review your provided information and click Create Project.

Create a project

Choose language setting

Enter basic information

Review and finish

Review and finish

Review the configurations you set for your project in the previous steps.

Projects in multiple languages allowed?

No

Language resource

LanguagestudioSUST

Azure Search resource

languagestudiosust-as2w2hjlhfwpzm

Project name

ABC

Description

--

Source language

English

Default answer when no response is returned

No answer found

Back

Next

Create project

Cancel

Next, From the **Manage sources** page select **Add source** > **URLS**.

Azure AI | Language Studio

Language Studio

Custom question answering

Azure Search

ABC

Manage sources

Edit knowledge base

Deploy knowledge base

Review suggestions

Project settings

Language Studio > Custom question answering > ABC - Manage sources

Manage sources

+ Add source

URLs

Files

Chitchat

Edit name

Refresh URL

Delete

ulster.ac.uk

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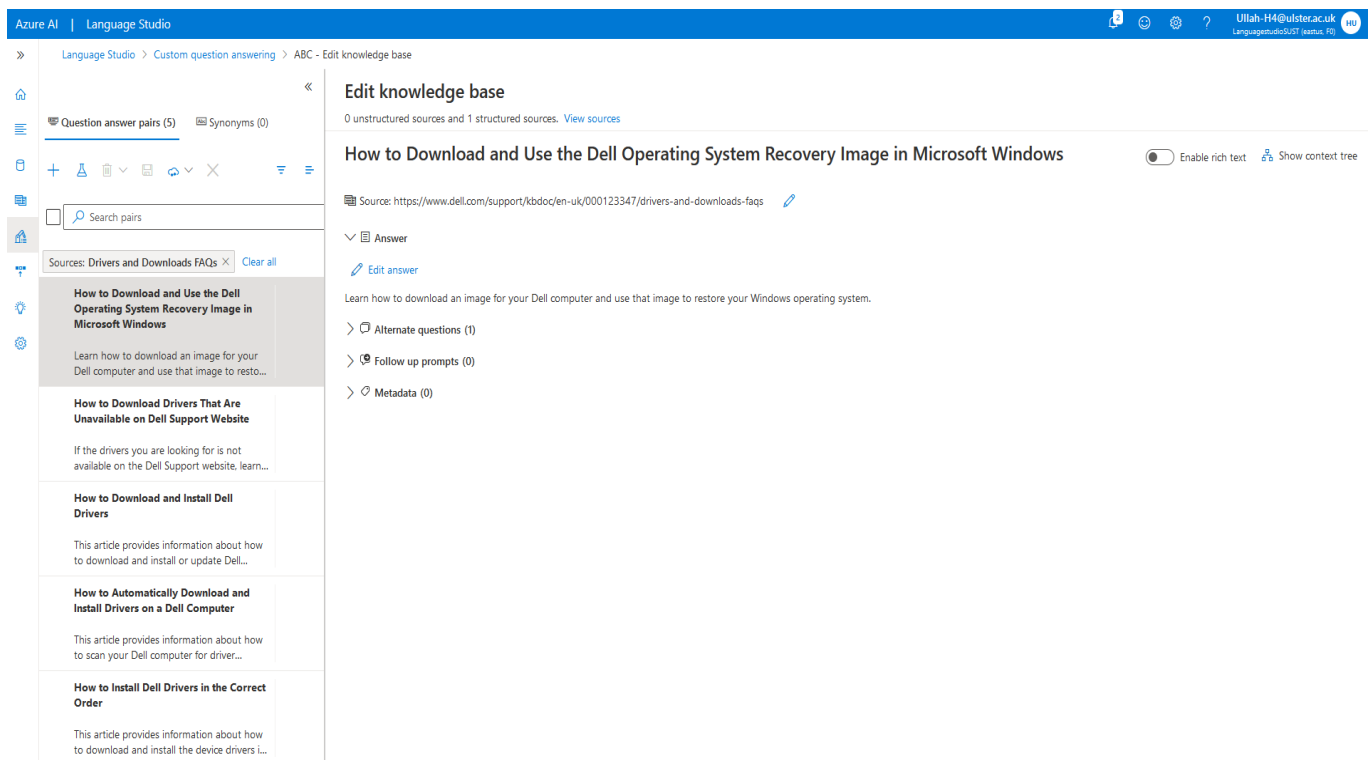
Select **Add url** enter the following values and then select **Add all**:

URL Name: Drivers and Downloads FAQs

URL Value: <https://www.dell.com/support/kbdoc/en-uk/000123347/drivers-and-downloads-faqs>

The extraction process takes a few moments to read the document and identify questions and answers. Question and answering will determine if the underlying content is structured or unstructured.

After successfully adding the source, you can then edit the source contents to add more custom question answer sets.



The screenshot shows the Azure AI Language Studio interface. The top navigation bar includes 'Azure AI | Language Studio' and a user profile 'Ullah-H4@ulster.ac.uk'. The main area is titled 'Edit knowledge base' and shows '0 unstructured sources and 1 structured sources'. A list of sources is displayed on the left, with the first source, 'How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows', selected. The right-hand pane shows the 'Answer' section for this source, displaying a generated response: 'Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.' Below the answer, there are sections for 'Alternate questions (1)', 'Follow up prompts (0)', and 'Metadata (0)'.

Test your project

1. Select the link to your source, this will open the edit project page.
2. Select **Test** from the menu bar > Enter the question **How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows?**. An answer will be generated based on the question answer pairs that were automatically identified and extracted from your source URL:

Azure AI | Language Studio

Language Studio > Custom question answering > ABC - Edit knowledge base

Edit knowledge base

0 unstructured sources and 1 structured sources. [View sources](#)

How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows

Source: <https://www.dell.com/support/kbdoc/en-uk/000123347/drivers-and-downloads-faqs>

Answer

Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.

> Alternate questions (1)

> Follow up prompts (0)

> Metadata (0)

Question answer pairs (5)

Search pairs

Sources: Drivers and Downloads FAQs x Clear all

- How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows**
Learn how to download an image for your Dell computer and use that image to resto...
- How to Download Drivers That Are Unavailable on Dell Support Website**
If the drivers you are looking for is not available on the Dell Support website, learn...
- How to Download and Install Dell Drivers**
This article provides information about how to download and install or update Dell...
- How to Automatically Download and Install Drivers on a Dell Computer**
This article provides information about how to scan your Dell computer for driver...
- How to Install Dell Drivers in the Correct Order**
This article provides information about how to download and install the drivers in the correct...

After the Question, you will see the following page:

We use optional cookies to improve your experience on our websites, such as through social media connections, and to display personalized advertising based on your online activity. If you reject optional cookies, only cookies necessary to provide you the services will be used. You may change your selection by clicking "Manage Cookies" at the bottom of the page. [Privacy](#)

Accept Reject Manage cookies

Azure AI | Language Studio

Language Studio > Custom question answering > ABC - Edit knowledge base

Edit knowledge base

0 unstructured sources and 1 structured sources. [View sources](#)

How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows

Source: <https://www.dell.com/support/kbdoc/en-uk/000123347/drivers-and-downloads-faqs>

Answer

Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.

> Alternate questions (1)

> Follow up prompts (0)

> Metadata (0)

Question answer pairs (5)

Search pairs

Sources: Drivers and Downloads FAQs x Clear all

- How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows**
Learn how to download an image for your Dell computer and use that image to resto...
- How to Download Drivers That Are Unavailable on Dell Support Website**
If the drivers you are looking for is not available on the Dell Support website, learn...
- How to Download and Install Dell Drivers**
This article provides information about how to download and install or update Dell...
- How to Automatically Download and Install Drivers on a Dell Computer**
This article provides information about how to scan your Dell computer for driver...
- How to Install Dell Drivers in the Correct Order**
This article provides information about how to download and install the drivers in the correct...

Test

Response options

☐ Include short answer response

☐ Use deployed knowledge base

[Show advanced options](#)

Clear

How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows

Inspect

Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.

If you check the box for **include short answer response** you will also see a precise answer, if available, along with the answer passage in the test pane when you ask a question.

3. Select **Inspect** to examine the response in more detail. The test window is used to test your changes to your project before deploying your project.

From the **Inspect** interface, you can see the level of confidence that this response will answer the question and directly edit a given question and answer response pair.

Test

Response options

☐ Include short answer response ⓘ

☐ Use deployed knowledge base ⓘ

[Show advanced options](#)

[Clear](#)

How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows

Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.

[Inspect](#)

Question

How to Download and Use the Dell Operating System Recovery Image in Microsoft Windows

Top answers

Select the most appropriate answer:

[Save this query as alternate question for this answer](#)

Answer [Edit question answer pair](#)

Learn how to download an image for your Dell computer and use that image to restore your Windows operating system.

Confidence score:
1.00

[See additional information](#)

Answer [Edit question answer pair](#)

This article provides information about how to download and install or update Dell drivers for your Dell desktop, laptop, all-in-one, workstation, printer, monitor, tablets, and so on.

Confidence score:
0.39

[See additional information](#)

Answer [Edit question answer pair](#)

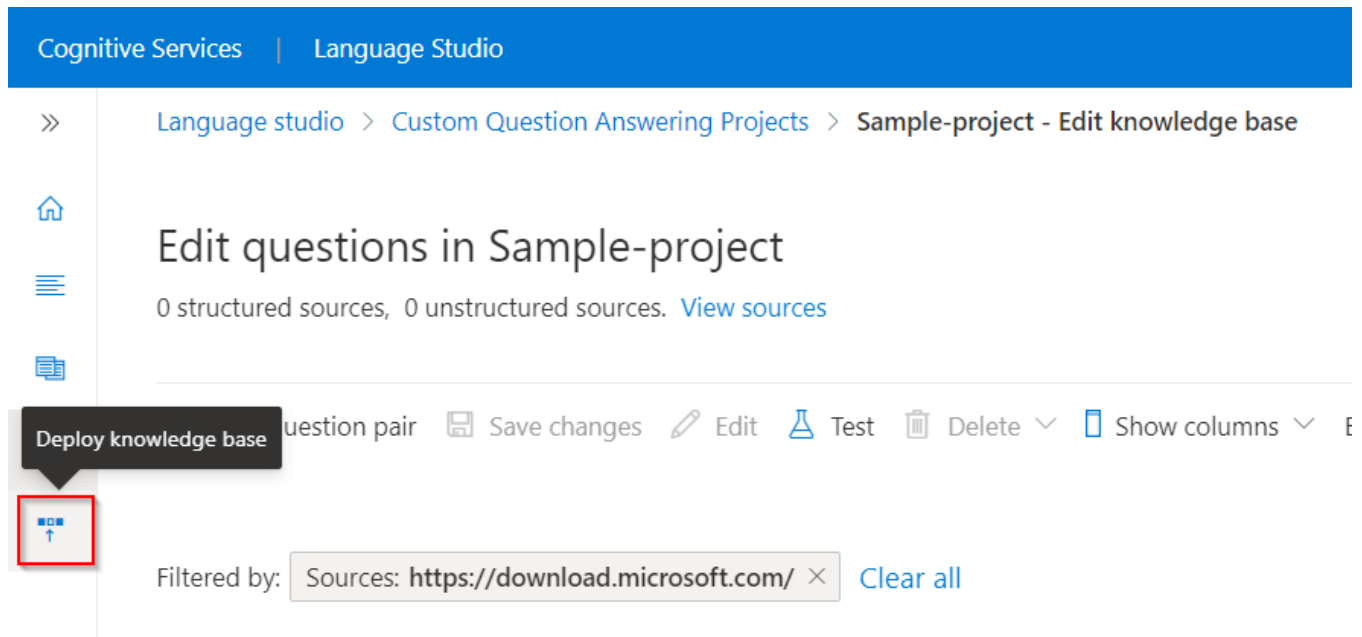
This article provides information about how to scan your Dell computer for driver updates, and automatically download and install the latest drivers.

Confidence score:
0.19

[See additional information](#)

Deploy your project

Select the Deploy project icon to enter the deploy project menu



When you deploy a project, the contents of your project move from the test index to a prod index in Azure Search.

Select **Deploy** > and then when prompted select **Deploy** again.

Deploy knowledge base

Deploy knowledge base and create a bot in a few clicks.

 Deploy  Get prediction URL

✔ Your knowledge base is now deployed. You can get can get your prediction URL or create a bot.

Knowledge base status

State: ⓘ	Deployed	✔	Resource:	Test-resource-01
Deployment Date: ⓘ	10/8/2021	✔	Location:	southcentralus
			Tier:	Standard (S)



Your project is now successfully deployed. You can use the endpoint to answer questions in your own custom application to answer or in a bot.

Clean up resources

If you will not continue to test custom question answering, you can delete the associated resource.